FYUGP B.Sc. PHYSICS HONOURS

PROPERTIES OF SOLIDS

Semester V · Major Elective (Materials Science)

MODULE III: SEMICONDUCTORS

Model questions on topics the past papers miss

Section A: 1 mark each, answer all

Intrinsic and extrinsic

1 What is the difference between an intrinsic and an extrinsic semiconductor? **(1 mark)**

Law of mass action

2 Write the law of mass action for a doped semiconductor, in terms of the electron and hole concentrations and the intrinsic concentration n_i. **(1 mark)**

Drift velocity

3 Write the drift velocity of a carrier in terms of its mobility μ and the applied field $\mathcal{E}$. **(1 mark)**

Conductivity of intrinsic semiconductors

4 Write the conductivity of an intrinsic semiconductor in terms of n_i, e and the electron and hole mobilities. **(1 mark)**

Section B: 2 marks each, answer all

Fermi level at 0 K

5 Where does the Fermi level of an intrinsic semiconductor lie at 0 K? Write the expression. **(2 marks)**

Electron concentration in the conduction band

6 Write the electron concentration in the conduction band in terms of the effective density of states N_c, E_c and E_F. **(2 marks)**

Donor ionisation energy

7 In an n-type semiconductor, what is the ionisation energy of the donors? Write it in terms of E_c and E_d. **(2 marks)**

Hole concentration in p-type

8 In a p-type semiconductor the ionised acceptor concentration is N_a^-. Write the hole concentration in terms of N_a^-. **(2 marks)**

Section C: 4 marks each, answer all

Conductivity from drift

9 An electron of mass m and charge e moves in a field $\mathcal{E}$ and has average time τ between collisions. Show that the conductivity of an electron gas of density n is $\sigma = ne^2\tau/m$. **(4 marks)**

Fermi level, n-type

10 Show that for an n-type semiconductor at moderate temperatures the Fermi level is $E_F = \dfrac{E_d + E_c}{2} + \dfrac{kT}{2}\ln\dfrac{N_d}{N_c}$. **(4 marks)**

Law of mass action

11 State the law of mass action and show that np depends only on the band gap and the temperature for an intrinsic semiconductor. **(4 marks)**

Fermi level, intrinsic

12 For an intrinsic semiconductor with effective masses m_n^* and m_p^*, show that the Fermi level is $E_F = \dfrac{E_c + E_v}{2} + \dfrac{3}{4}kT\ln\dfrac{m_p^*}{m_n^*}$. **(4 marks)**

Fermi level and temperature

13 Explain the variation of the Fermi level with temperature for an n-type semiconductor, and say what happens at high temperature. **(4 marks)**

Temperature dependence of conductivity

14 Sketch $\ln\sigma$ against $1/T$ for an intrinsic semiconductor. Show that the slope gives the band gap E_g. **(4 marks)**

Section D: 10 marks each, answer any ONE

Electron concentration, derivation

15 Derive the electron concentration in the conduction band of an intrinsic semiconductor, and write the effective density of states N_c in terms of m_n^*, T and h. **(10 marks)**

Hole concentration and n_i, derivation

16 Derive the hole concentration in the valence band, and the intrinsic carrier concentration $n_i = 2(2\pi kT/h^2)^{3/2}(m_n^* m_p^*)^{3/4}\exp(-E_g/2kT)$. **(10 marks)**

Conductivity of n-type

17 Explain how the conductivity of an n-type semiconductor varies with temperature in three regions (ionisation, saturation and intrinsic), and derive $\sigma = eN_d\mu_n$ for the middle region. **(10 marks)**

MODULE III: MODEL ANSWERS

For checking your work. Each answer shows the key step and the result.

Section A. 1. An intrinsic semiconductor is pure, with $n = p = n_i$. An extrinsic semiconductor is doped with donors or acceptors, so one carrier type dominates (book §7.1 to §7.3). **2.** $np = n_i^2$ (book Eq. 7.34). **3.** $v_d = \mu\mathcal{E}$ (book Eq. 7.3). **4.** $\sigma = en_i(\mu_n + \mu_p)$ (book Eq. 7.9).

Section B. 5. $E_F = E_i = (E_c + E_v)/2$ (book Eq. 7.31). **6.** $n = N_c \exp[-(E_c - E_F)/kT]$ (book Eq. 7.20). **7.** $\Delta E = E_c - E_d$ (book §7.6a; Eq. 7.44). **8.** $p \cong N_a^-$ (book Eq. 7.46).

Section C. 9. $J = nev_d$ and $v_d = \mu\mathcal{E}$ with $\mu = e\tau/m$ give $J = (ne^2\tau/m)\mathcal{E}$; compare with $J = \sigma\mathcal{E}$ (book Eqs. 7.3 to 7.6). **10.** Use $n = N_d^+$ (Eq. 7.38) and $N_d^+ = N_d \exp[-(E_F - E_d)/kT]$ (Eq. 7.40) with $n = N_c \exp[-(E_c - E_F)/kT]$ (Eq. 7.20); take logarithms (Eq. 7.41). **11.** $np = n_i^2 = N_c N_v \exp(-E_g/kT)$ (Eqs. 7.32 to 7.34), a constant at fixed T. Its value depends only on E_g and T for an intrinsic material. **12.** Equate $N_c \exp[-(E_c - E_F)/kT] = N_v \exp[-(E_F - E_v)/kT]$ (Eqs. 7.20, 7.26), take logarithms, and use N_c, N_v (Eqs. 7.21, 7.27) (Eq. 7.30). **13.** Rises with temperature across the ionisation region to E_F near E_d, then falls towards E_i as T rises; the formula (7.41) fails at $T = 0$ and at very high T (book §7.6a, Fig. 7.8). **14.** From $\sigma = CT^{3/2} \exp(-E_g/2kT)$ (Eq. 7.11), $\ln\sigma = -(E_g/2k)(1/T) + \frac{3}{2}\ln T +$ const (Eq. 7.12). The slope is $-E_g/2k$, so $E_g = -2k \times$slope (book Fig. 7.5).

Section D. 15. Integrate $dn = D(E)f(E)dE$ with D and f (Eqs. 7.13 to 7.18), use $kT \ll E - E_F$ and $x = (E - E_c)/kT$ (book §7.5.1); $n = N_c \exp[-(E_c - E_F)/kT]$ with $N_c = 2(2\pi m_n^* kT/h^2)^{3/2}$ (Eqs. 7.19 to 7.21). **16.** Integrate $dp = D(E)[1 - f(E)]dE$ over $E17. Ionisation region: σ rises as donors ionise (point A to B). Saturation region: σ falls with the mobility (B to C). Intrinsic region: σ rises as n_i grows (C to D). Middle region: $n \cong N_d^+$ (Eq. 7.38), so $\sigma = en\mu_n \cong eN_d\mu_n$ (Eq. 7.45).

Mapping to the syllabus and textbook

Syllabus item	Topic	Puri and Babbar section	Model questions
12	Intrinsic and extrinsic semiconductors	7.1 to 7.3	1, 2, 7, 8
13	Drift velocity	7.4	3, 9
14	Mobility and conductivity of intrinsic semiconductors	7.4 (Eqs. 7.3 to 7.12)	4, 14
15	Carrier concentration and Fermi level	7.5, 7.6	5, 6, 10, 11, 12, 13, 15, 16
16	Conductivity for intrinsic and extrinsic semiconductors	7.4, 7.6 (Eqs. 7.45, 7.51)	17

Questions are drafted for the gaps in the past papers (not from any past paper). Chapter questions and problems of the book are not used.